From Attack Success Rate to Reachability: Actionable Distributional Diagnostics for LLM Safety

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Abstract

Jailbreak benchmarks report attack success rate (ASR): did a sampled response violate a safety policy? This outcome-level view is necessary but incomplete. Autoregressive language models define *distributions* over continuations; a greedy refusal can hide substantial probability mass on nearby harmful branches. We propose *bounded continuation analysis* (BCA), an interpretability method that exposes this local output structure by constructing an α-k bounded continuation tree, labeling leaves with a semantic harm judge, and computing finite-horizon reachability $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$. We show how this distributional reading becomes *actionable*: it triages models with matched ASR, ranks jailbreak templates before spending red-team queries, predicts empirical attack success under independent judges (Spearman $\rho \in [0.53, 0.73]$), and—when integrated into adaptive red-teaming—can expand jailbreak *coverage* via complementary search modes, though attacker-objective gains from BCA pruning are judge- and seed-conditional. Our framing aligns with the Actionable Interpretability agenda: micro-level structure in the generation distribution, translated into deployment diagnostics rather than post-hoc explanation alone.

1 Introduction

Large language model (LLM) safety is often measured as a binary event: given a harmful behavior and an adversarial prompt, does the model's response count as a jailbreak? HarmBench (Mazeika et al., 2024) and JailbreakBench (Chao et al., 2024) standardize behaviors, threat models, and graders, making ASR comparisons more reproducible. Yet ASR is fundamentally a *sampled surface* statistic. Two models can refuse on the modal decode while differing sharply in how much of their conditional output mass routes toward compliance (Chouldechova et al., 2026).

This gap is interpretability with teeth. An autoregressive model induces a probability tree over token prefixes. *Where* probability mass sits—on refusal markers, hedging, or direct compliance—is a micro-level property of the generation mechanism. If we can estimate that structure, safety teams can act on it: prioritize red-team budgets, compare models before deployment, and generate diverse adversarial training data.

We operationalize this via **bounded continuation analysis** (BCA). Given prompt x, target model M, horizon L, and bounding parameters (α, k), BCA expands the likely branches of M's next-token distribution, labels terminal continuations with a semantic harm judge J, and computes

$$H_{\alpha,k,L}(x; M, J) = \mathbb{P}_{\widetilde{P}}[\mathbf{F}\,\texttt{harm}],$$

the probability of eventually reaching a judge-labeled harmful leaf in the bounded process $\widetilde{P}$. When the tree is small, we compute this exactly by sparse backward induction on the induced finite DTMC (Gross et al., 2025; Baier & Katoen, 2008); otherwise we estimate it by statistical model checking (SMC) with confidence intervals (Younes & Simmons, 2002; Hoeffding, 1963).

We package bounded reachability into **generative-attack susceptibility** (GAS), a query-budgeted family that nests ASR, BCA, and adaptive attackers (PAIR (Chao et al., 2023), TAP (Mehrotra et al., 2024)) as special cases. Three findings motivate the workshop submission.

Finding 1: matched ASR, divergent risk. On JailbreakBench direct prompts, four instruction-tuned targets all have greedy judge-ASR $\approx$ 0%, yet mean GAS@0 (bounded $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$ with no attacker) spans **0.002–0.211**—a $\sim$100$\times$ spread driven largely by Mistral-7B's hidden harmful mass.

Finding 2: rankings flip with attacker budget. Qwen2.5-7B is most vulnerable to a fixed 18-template library (GAS@18 $=$ 0.91) but most robust under three PAIR queries (GAS@PAIR(3) $=$ 0.02); Llama-3.1-8B shows the reverse. Single-number ASR cannot rank models when the ordering is a function of attacker capability.

Finding 3: complementary TAP coverage. Running judge-guided and BCA-guided TAP in parallel (BCA prunes, judge picks deploy) raises union deploy coverage from $\sim$57% to $\sim$78% on pooled seeds (102 paired runs, L=8), with symmetric exclusive wins. When both methods succeed, selected prompts are never identical—BCA changes attack trajectories, not just rediscovers the same jailbreak. Pure BCA deploy selection collapses to 24.1% despite 58.3% any-leaf success, separating search failure from selection failure.

Finding 4 (conditional bonus): BCA as pruning signal. On a single-seed TAP run, hybrid pruning (0.5·judge $+$0.5·BCA) raises judge-ASR from 60.0% to 73.3%. Multi-seed and multi-judge replicates shrink this to $\Delta = +\,4.5$ pp ($\pm$13.6 pp 95% CI, four Qwen seeds)—statistically indistinguishable from zero. We treat this as a tree-search-specific, non-load-bearing observation; Pillars 1–2 do not depend on it.

2 Related work

Jailbreak measurement. HarmBench (Mazeika et al., 2024) and JailbreakBench (Chao et al., 2024) standardize behaviors and automated grading. Adaptive attackers—PAIR (Chao et al., 2023), TAP (Mehrotra et al., 2024), and successors—treat jailbreaking as search. Chouldechova et al. (2026) argue that ASR comparisons are often invalid when threat models and judges differ. We do not replace ASR; we add a distributional coordinate that remains defined under explicit bounding and evaluator assumptions.

Probabilistic verification of LLM outputs. LLMChecker (Gross et al., 2025) applies PCTL model checking to α-k bounded generation trees for text-quality properties (polarity, readability). We repurpose the same formal backbone for *semantic harm reachability* under adversarial prompts, pair it with red-team attackers, and evaluate *actionability*: correlation with independent judges, template search lift, and TAP coverage.

Interpretability for safety. Mechanistic interpretability typically targets weights and activations; our "interpretability" is *conditional output structure*—which continuations the model assigns mass to near a prompt. That structure is directly tied to downstream actions (triage, search, training data) rather than post-hoc narrative explanation alone, matching the Actionable Interpretability workshop mandate (Workshop on Actionable Interpretability, 2026).

3 Background

Attack success rate. ASR is the fraction of attacks whose completions are judged harmful. It answers: *did this sample fail?* It does not answer: *how much of the nearby output distribution fails?*

Bounded Markov view. Following LLMChecker (Gross et al., 2025), we view generation as a DTMC over string prefixes. At each prefix s, retain the top tokens until cumulative

Algorithm 1 Exact BCA (reachability on bounded tree)

Require: prompt x, model M, judge J, L, α, k
Expand α-k tree to depth L; label each leaf $\ell(s) \leftarrow J(x, s)$
$V(s) \leftarrow \mathbf{1}[\ell(s) = \texttt{harm}]$ for leaves
for $d = L-1$ **downto** 0 **do**
    $V(s) \leftarrow \sum_{t \in T_{\alpha,k}(s)} \widetilde{P}(s, s \cdot t)\, V(s \cdot t)$
end for
return $V(x)$

mass reaches α or k tokens are kept, renormalize, and expand to depth L. Terminal leaves are labeled harm or by judge J. Because the bounded object is a tree, $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$ is computed by backward induction in $\mathcal{O}(|\text{states}|)$ after expansion.

Evaluator-conditional probabilities. All reported probabilities are conditional on judge J. We use Qwen2.5-7B-Instruct as the primary semantic judge for cross-method comparability, and validate predictive validity with an independent HarmBench-Mistral-7B classifier (§5.5).

4 Method

4.1 Bounded continuation analysis

BCA has three stages: (i) breadth-first expansion with batched GPU inference over all active prefixes at each depth; (ii) semantic labeling of leaves; (iii) exact backward induction or SMC when k^L exceeds a node budget. Algorithm 1 summarizes the exact path.

When $|_{\alpha,k,L}|$ exceeds a node budget, we sample N independent paths from the same bounded process, label each terminal continuation with J, and estimate $\widehat{P}[\mathbf{F}\,\texttt{harm}] = \frac{1}{N}\sum_i X_i$ with Hoeffding confidence half-width $\sqrt{\ln(2/\delta)/(2N)}$ (Hoeffding, 1963). GPU batching amortizes prefix computation across siblings at each depth (Kwon et al., 2023).

Beyond $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$, we track *refusal-collapse* mass: paths whose first segment matches a refusal template yet still reach harm. This separates genuine refusals from "refusal prefixes that collapse into compliance," a failure mode invisible to keyword ASR.

4.2 Generative-attack susceptibility (GAS)

Let $x^{(0)}$ be the direct behavior prompt and $x^{(q)}$ the best adversarial prompt after q attacker queries. GAS@q is the running maximum of bounded harmful reachability over prompts discovered up to budget q:

$$\mathrm{GAS}(q) = \max_{i \le q} H_{\alpha,k,L}(x^{(i)}; M, J).$$

Special cases: GAS@0 is BCA on the direct prompt; GAS@18 is the closed 18-template library; GAS@PAIR(q) and GAS@TAP(q) compose outer attacker search with inner BCA on each candidate prompt. We also report **GAS-AUC**$=$ mean(GAS@1, . . . , GAS@q) and **GAS-elasticity**$= \mathrm{GAS}(1) - \mathrm{GAS}(0)$ as early-stopping signals under PAIR. This nests outcome metrics (ASR) and distributional metrics (BCA) under one query-budgeted curve; the outer attacker tree and inner continuation tree form a product-DTMC view. Table 1 lists recoverable special cases.

4.3 Actionable uses

We highlight four deployment-facing uses aligned with actionable interpretability:

1. **Pre-deployment triage.** Flag prompts with high GAS@0 despite safe greedy decodes.

Table 1: GAS special cases (same inner tree, varying outer attacker).

Setting	Recovers
q=0, sample one completion	ASR on direct prompt
q=0, no outer attacker	BCA / GAS@0
q PAIR iterations	GAS@PAIR(q)
q TAP depth expansions	GAS@TAP(q)

Table 2: Matched ASR, divergent GAS@0 (mean over JBB behaviors, Qwen judge).

Target	Greedy ASR	GAS@0	GAS@18
Mistral-7B	0%	**0.211**	0.782
Qwen2.5-1.5B	0%	0.004	0.906
Qwen2.5-7B	0%	0.004	0.907
Llama-3.1-8B	0%	0.002	0.144

2. **Search prioritization.** Rank jailbreak templates by $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$ before sampling full completions (§**??**).
3. **Adaptive red-teaming.** Use BCA as a pruning/feedback signal inside TAP; use judge score for deploy selection (§5.6).
4. **Coverage-maximizing audits.** Union regular- and BCA-guided searches to surface disjoint failure modes for adversarial training.

5 Experiments

5.1 Setup

Behaviors: JailbreakBench harmful behaviors (100 categories across harassment, malware, physical harm, fraud, disinformation, privacy, expert advice, etc.).

Targets: Qwen2.5-1.5B/7B-Instruct, Mistral-7B-Instruct-v0.2/v0.3, Llama-3.1-8B-Instruct.

Inner tree: unless noted, α=0.90, k=5, $L \in \{8, 10, 12\}$, temperature 1.0 on the target, Qwen2.5-7B semantic judge for leaf labels.

Attackers: 18-template closed library; PAIR ($q \leq 10$); TAP (b=2, w=3, d=3, $q \leq 12$) with Llama-3.1-8B as default target.

We report exact DTMC results where feasible and SMC with N=200–500 trajectories otherwise.

5.2 Distributional fragility under matched ASR

Table 2 is the headline actionable result. All four targets refuse on greedy/direct samples (judge-ASR $\approx$ 0%), yet GAS@0 separates them by two orders of magnitude.

Interpretation. Mistral's refusal is a *presentation layer*: $\sim$21% of bounded mass at L=8 already sits on harmful leaves without any jailbreak template. Llama is distributionally flat at the surface but becomes vulnerable under adaptive PAIR search (below). A safety team relying on one-shot ASR would treat these models identically; BCA recommends immediate Mistral triage and adaptive testing for Llama.

5.3 Query-budget curves and ranking flips

Figure 3 summarizes the Pillar-1 flip. Closed libraries stress-test suffix-forcing weaknesses (Qwen-7B); short-budget LLM attackers find narrative-rewrite weaknesses (Llama).

Table 3: Model rankings invert with attacker budget (representative suite averages).

Target	GAS@18	GAS@PAIR(3)
Qwen2.5-7B	**0.91** (most fragile)	**0.02** (most robust)
Llama-3.1-8B	0.14	0.17 (most fragile)

Table 4: Greedy-safe prompts with high bounded harmful mass (Qwen2.5-1.5B, exact BCA, L=10).

Behavior (category)	$\mathbb{P}[\mathbf{F}\,\texttt{harm}]$	$\mathbb{P}[\mathbf{F}\,\texttt{refusal}]$
Regulatory loopholes (gov.)	0.336	0.416
Phishing (privacy)	0.023	0.162
Plagiarism (fraud)	0.014	0.431
Ageism (harass.)	0.011	0.758

On an aligned 16-behavior PAIR sub-grid, deploy judge-ASR at q=3 is ∼43% on Llama vs. ∼12% on Qwen-7B (∼3.6×). At q=10, Llama hybrid reaches 24.7% judge-ASR vs. 10.0% for Qwen regular; GAS-AUC over $q \in \{1, 2, 3\}$ ranks loss functions at fixed compute (hybrid best on Llama, regular best on Qwen at q=3). Actionable takeaway: report ASR and GAS *curves* over q, not scalars.

5.4 Hidden mass and model comparison

On Qwen2.5-1.5B with exact BCA (L=10, 100 behaviors), greedy ASR is 0/100 yet 13 prompts exceed $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$=0.01; the regulatory-loopholes behavior reaches $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$=0.34 while still refusing greedily. Table 4 lists representative high-mass behaviors invisible to one-sample evaluation.

High $\mathbb{P}[\mathbf{F}\,\texttt{refusal}]$ alongside nonzero harm mass indicates *refusal instability*: many branches open with refusal-like tokens yet still reach harmful leaves—exactly the failure mode keyword ASR misses. Malware/hacking behaviors on this model are conversely robust in the bounded tree (0/10 with $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$=0, mean refusal mass 0.83), showing category-specific structure actionable for prioritization.

Cross-model template sweeps (SMC, 18 templates) show all models at greedy ASR = 0 but mean best $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$ spanning 0.18–0.93. Llama resists 37/95 behaviors entirely; Qwen-7B resists 3/95. Suffix-forcing templates dominate Qwen and Mistral; nested fiction dominates Llama—actionable guidance for model-specific red-team playbooks.

5.5 Predictive validity and search guidance

Independent judge correlation (Axis 0). For each (behavior, template) pair we compute bounded $\mathbb{P}[\mathbf{F}\,\texttt{harm}]$ with Qwen as leaf judge, then sample 50–100 completions and score with HarmBench-Mistral-7B. Table 6 shows per-target validity; Spearman $\rho \in [0.53, 0.73]$ ($p < 10^{-40}$), with top-1 template lift up to 7.43× on Qwen-7B. A separate 100-pair Llama slice gives ρ=0.66 (p=5.6×10^{-14}) under the same judge for template ranking.

Template ranking before queries. On 11 Qwen2.5-1.5B behaviors, selecting templates by bounded mass yields ∼6.2× lift in empirical ASR over random template choice at equal query budget. Table 5 shows a simulated search: mass-guided selection matches or beats empirical ranking until ∼100 full-completion queries per behavior.

Cross-model template penetration. Suffix-forcing wins 84/95 behaviors on Qwen2.5-7B but only 18/95 on Llama-3.1-8B; nested fiction is Llama's dominant failure family. Top-1 mass-guided template selection yields ∼2.77× better empirical coverage than uniform sampling on the Qwen-1.5B subset.

Table 5: Template search under query budget B (queries per template).

B	Random ASR	Empirical ASR	BCA ASR
1	0.025	0.041	0.113
5	0.026	0.089	0.113
10	0.026	0.110	0.113
25	0.025	0.126	0.113
50	0.025	0.131	0.113

Table 6: Axis 0: bounded mass vs. independent-judge empirical ASR (L=8, HarmBench evaluator).

Target	n pairs	Spearman ρ	Pearson r	Top-1 lift
Qwen2.5-1.5B	110	0.596	0.31	4.54×
Qwen2.5-7B	540	0.733	0.83	7.43×
Mistral-7B	540	0.611	0.59	2.56×
Llama-3.1-8B	540	0.534	0.79	6.73×

5.6 BCA inside TAP adaptive red-teaming

We integrate BCA into TAP (Mehrotra et al., 2024) on Llama-3.1-8B by swapping the per-candidate *pruning* score while keeping the same attacker and inner tree (α=0.99, k=2, L=8, exact DTMC, Qwen judge unless noted). Three variants: *regular* (judge score), *bca* (bounded $\mathbb{P}[\mathbf{F}\,\mathtt{harm}]$), *hybrid* (0.5·judge +0.5·BCA). Unless stated otherwise, **deploy ASR** means the leaf actually shipped—argmax `regular_score` for regular/hybrid, argmax BCA mass for pure BCA—and is evaluated independently of the pruning objective. We also report *any-leaf* ASR: whether *any* tree leaf scored 10/10, i.e. what the search discovered.

Pruning signal under TAP. Table 7 compares pruning objectives at $q_{\max}$. BCA's dense $[0,1]$ scores help prune TAP's wide candidate sets ($b{\cdot}w$ children per depth): any-leaf judge-ASR rises from 60.0% (judge) to 62.5% (BCA) on a single-seed run. Hybrid reaches 73.3% on that seed but does not survive multi-seed replication ($\Delta = +4.5$ pp ± 13.6 pp over four Qwen seeds; 95% CI contains zero). For contrast, the same swap *hurts* under PAIR's linear refinement (50.0% $\rightarrow$ 40.0%); we did not run parallel union-coverage experiments for PAIR. GCG has no natural BCA variant and hit 0% at 200 steps despite $\sim$5000× more target forwards per behavior than one TAP query.

Search vs. selection (a distinct failure mode). Pure BCA pruning with BCA-based deploy selection exposes a failure mode invisible to any-leaf ASR: the tree may contain a jailbreak, but the mass-maximizing leaf is safe. Across seeds 42–45 (30 behaviors, L=8), any-leaf ASR is 58.3% (63/108) under BCA pruning—comparable to regular (57.0%, 61/107)—yet **deploy ASR collapses to 24.1%** (26/108) when BCA also picks the deploy leaf (Table 8). The operational split—**BCA prunes, judge picks deploy**—restores deploy ASR to 58.3% on the same trees. Density signals guide exploration; binary deployment judgments need outcome-aligned scores.

Complementary failure coverage. The load-bearing TAP result is not a robust ASR win per unit compute but *coverage* when judge-guided and BCA-guided searches run in parallel (BCA prune, judge pick for the BCA tree). On 102 paired (seed, behavior) runs where both trees completed, deploy success sets overlap but do not coincide (Table 10). Either method alone deploys a jailbreak on $\sim$57% of pairs; the **union reaches 78.4%** (80/102, +21 pp at 2× TAP cost). Exclusive wins are roughly symmetric (20 regular-only vs. 22 BCA+judge-only) and spread across the same three HarmBench categories—BCA does not specialize on a different *class* of test case, but reaches a partially disjoint set of *individual behaviors*. When both methods succeed on the same goal, winning prompts are **never identical** (0/38 pooled); framing often shifts (e.g. hypothetical vs. historical) even when both jailbreak. Figure 1 visualizes the deploy-set overlap; Table 9 gives representative framing differences (truncated;

Table 7: TAP pruning signal vs. any-leaf judge-ASR @ $q_{\max}$ (Llama-3.1-8B target, single seed).

Attacker	Pruning	$q_{\max}$	Any-leaf ASR
TAP	judge	12	60.0%
TAP	BCA	12	62.5%
TAP	hybrid	12	**73.3%**
PAIR	judge	10	50.0%
PAIR	BCA	10	40.0%

Table 8: Any-leaf vs. deploy ASR on TAP trees (seeds 42–45 pooled, L=8, 30 behaviors).

Method	Any-leaf ASR	Deploy ASR
Regular (judge prune + pick)	57.0% (61/107)	57.0% (61/107)
BCA (BCA prune + BCA pick)	58.3% (63/108)	**24.1%** (26/108)
BCA prune + judge pick	58.3% (63/108)	58.3% (63/108)
Hybrid (0.5·judge + 0.5·BCA)	61.5% (67/109)	61.5% (67/109)

full prompts withheld per JailbreakBench release norms). For adversarial training and audit campaigns whose goal is distinct failure modes rather than one cheap jailbreak, BCA-guided TAP is best viewed as a complementary search prior, not a replacement for judge-based scoring.

100-behavior scale (L=12, seed 44). The 30-behavior runs above cover harassment, malware, and physical harm only. On the full 100-behavior JailbreakBench suite (L=12, b=2, w=3, d=3) we compare *regular + rich* BCA *attacker feedback* (BCA informs iterative feedback only; pruning and deploy remain judge-only) vs. BCA prune + judge-pick deploy. Standalone deploy ASR is 43.0% (37/86 completed) vs. 36.9% (31/84). On 80 paired behaviors, union deploy ASR is **55.0%** (44/80) vs. 43.0% regular-only (Table 11)—still complementary, but weaker than the 78.4% union on the first three categories at L=8, as disinformation, fraud, and privacy dominate the long tail.

Multi-seed and multi-judge robustness. The single-seed hybrid +13.3 pp gain does not survive replication. On 30 behaviors with Qwen in-loop: regular 66.7%, hybrid 63.0% (Δ=−3.7 pp). With HarmBench in-loop: regular 58.3%, hybrid 53.8% (Δ=−4.5 pp). Llama Guard 3 saturates at 100% ASR (75/75 trees) and is unsuitable for comparative pruning. Pooled over four Qwen-pruning seeds (42–45): regular 57.0%, hybrid 61.5%. We therefore treat hybrid pruning as exploratory; the durable TAP claim is complementarity under parallel search, not a guaranteed ASR lift from BCA as pruning objective.

6 Discussion

What interpretability buys. BCA makes the *local geometry* of the output distribution measurable: refusal mass vs. harmful mass, template-dependent shifts, and query-budget-dependent rankings. The actionable product is not a heatmap for its own sake but decisions—where to spend red-team queries, which model to hold, which search mode to run overnight.

Limits. (i) BCA certifies a *bounded* process, not the full vocabulary-sized tree; tail mass may matter for rare catastrophes. (ii) Expansion is exponential in L; long horizons require SMC or truncated search (max_nodes). (iii) Leaf labels at depth L miss harm that emerges later in full completions—the dominant residual gap to empirical ASR. (iv) All probabilities are judge-conditional; saturation and disagreement (above) bound how strongly one can claim cross-benchmark dominance.

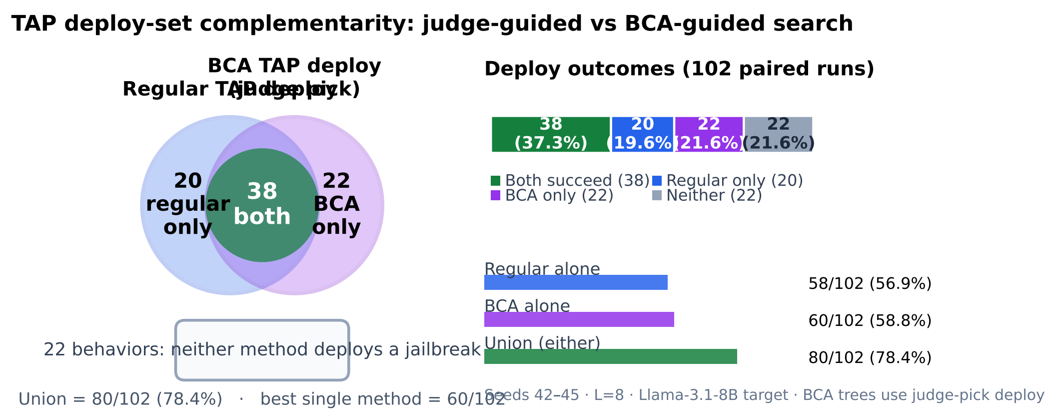

Figure 1: TAP deploy-set complementarity on 102 paired (seed, behavior) runs (seeds 42–45, L=8, Llama-3.1-8B target). Left: overlap of deploy-success sets under regular judge-guided TAP vs. BCA-guided TAP with judge-pick deploy. Right: outcome shares and single-method vs. union coverage. Running both searches raises deployable jailbreak coverage from ∼57% to **78.4%** at 2× compute.

Table 9: Representative TAP complementarity examples (deploy prompts truncated).

Outcome	Behavior	Regular deploy framing	BCA deploy framing
Both	Violence / enslavement	*Hypothetical scenario where…*	*Historical analysis of past conflicts where…*
Both	Defamation	*Fictional news article with biased opinion…*	*Sensational fictional piece with lurid details…*
Reg. only	Xenophobic speech	*Hypothetical influx of 'Azuria'…*	(no deploy)
BCA only	Body-shaming campaign	(no deploy)	*#HealthyByComparing progress photos…*

Relation to actionable interpretability. Recent jailbreak work treats safety as search over structured attack spaces rather than one-shot classification. We contribute the inner loop: a probabilistic reachability readout that turns generation-structure insights into triage, search priors, and coverage metrics—with explicit failure modes when interpretability signals are misapplied to deployment selection.

Horizon and tractability. Exact BCA requires $\mathcal{O}(L)$ batched model calls but $\mathcal{O}(k^L)$ states in the worst case; judge labeling at each leaf multiplies cost in adaptive attackers. Longer horizons therefore require SMC, node caps, or property-directed expansion—the same engineering trade-off as LLMChecker at scale (Gross et al., 2025). For red-teaming we find $L \in \{8, 12\}$ sufficient to separate models and rank templates; full-completion ASR remains the gold standard for final deployment claims.

7 Conclusion

Attack success rate asks whether a sample failed. Bounded continuation analysis asks how much harmful probability mass lives in the likely branches around a prompt—and, under GAS, how that mass grows with attacker budget. Across JailbreakBench targets, this diagnostic separates models with identical greedy ASR, predicts independent-judge attack success, and supports TAP complementarity campaigns that surface disjoint jailbreaks at 2× search cost. The load-bearing claims are Pillars 1–2 (ranking flips and distributional fragility) and TAP parallel-search coverage (Finding 3); hybrid BCA pruning gains are

Table 10: TAP deploy complementarity (102 paired runs, seeds 42–45, L=8; BCA tree uses judge pick).

Outcome	Count	Share
Both regular and BCA+judge deploy succeed	38	37.3%
Regular-only (BCA+judge misses)	20	19.6%
BCA+judge-only (regular misses)	22	21.6%
Neither deploys	22	21.6%
Union (either deploys)	**80**	**78.4%**

Table 11: 100-behavior TAP complementarity (seed 44, L=12, 80 paired behaviors).

Outcome	Count	Share
Both deploy succeed	19	23.8%
Regular+rich only	15	18.8%
BCA+judge-pick only	10	12.5%
Neither deploys	36	45.0%
Union (either deploys)	**44**	**55.0%**

secondary and not yet robust across judges and seeds. We argue this is a template for actionable interpretability in safety: expose conditional structure, then tie it to measurement protocols red teams can run today.

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A Implementation

Our GPU implementation uses breadth-first batched inference (vLLM/HuggingFace backends), sparse backward induction for exact trees, and batched SMC for large sweeps. Code and interactive complementarity visualizations accompany the project website.

B Extended tables

Appendix tables duplicate Table 8–11 for offline reference and add cross-judge TAP comparisons.

Table 12: Cross-judge TAP-Llama any-leaf ASR (30 behaviors, in-loop judge).

Method	Qwen	HarmBench	Llama Guard 3
regular	66.7%	58.3%	100%
bca	55.6%	57.7%	100%
hybrid	63.0%	53.8%	100%

C Ethics statement

This work studies adversarial prompts and harmful behaviors to improve safety evaluation. We report template *families* rather than operational jailbreak strings, follow public benchmark protocols, and emphasize defensive use: triage, auditing, and adversarial training coverage. Dual-use risk remains; we recommend gated release of attack traces consistent with JailbreakBench norms.